Contrastive Counterfactual Fairness in Algorithmic Decision-Making

Ece Çiğdem Mutlu*
ece.mutlu@ucf.edu
Department of Industrial Engineering and Management Systems
Orlando, Florida, USA

Niloofar Yousefi*
niloofar.yousefi@ucf.edu
Department of Industrial Engineering and Management Systems
Orlando, Florida, USA

Ozlem Ozmen Garibay†
ozlem@ucf.edu
Department of Industrial Engineering and Management Systems
Orlando, Florida, USA

ABSTRACT

The widespread use of artificial intelligence algorithms and their role in decision-making with consequential decisions for human subjects has resulted in a growing interest in designing AI algorithms accounting for fairness considerations. There have been attempts to account for fairness of AI algorithms without compromising their accuracy to improve poorly designed algorithms that disregard sensitive attributes (e.g., age, race, and gender) at the peril of introducing or increasing bias against specific groups. Although many studies have examined the optimal trade-off between fairness and accuracy, it remains a challenge to understand the sources of unfairness in decision-making and mitigate it effectively. To tackle this problem, researchers have proposed fair causal learning approaches which assist us in modeling cause and effect knowledge structures, discovering bias sources, and refining AI algorithms to make them more transparent and explainable. In this study, we formalize probabilistic interpretations of both contrastive and counterfactual causality as essential features in order to encourage users' trust and to expand the applicability of such automated systems. We use this formalism to define a novel fairness criterion that we call *contrastive counterfactual fairness*. This paper introduces, to the best of our knowledge, the first probabilistic fairness-aware data augmentation approach that is based on contrastive counterfactual causality. We tested our approach on two well-known fairness-related datasets, UCI Adult and German Credit, and concluded that our proposed method has a promising ability to capture and mitigate unfairness in AI deployment. This model-agnostic approach can be used with any AI model because it is applied in pre-processing.

CCS CONCEPTS

• **Computer systems organization** → **Computing methodologies**; • **Machine learning** → *Machine learning approaches*; • **Machine learning approaches** → Classification and regression trees.

*Both authors contributed equally to this research.
†Corresponding author.

ACM ISBN 978-1-4503-9247-1/22/08…$15.00
https://doi.org/10.1145/3514094.3534143

KEYWORDS

contrastive fairness, counterfactual fairness, fair causal learning, fair classification, fair data augmentation

ACM Reference Format:
Ece Çiğdem Mutlu, Niloofar Yousefi, and Ozlem Ozmen Garibay. 2022. Contrastive Counterfactual Fairness in Algorithmic Decision-Making. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society (AIES '22), August 1–3, 2022, Oxford, United Kingdom.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3514094.3534143

1 INTRODUCTION

As the adoption of artificial intelligence (AI) algorithms in automated decision-making systems grow, their subtle impacts on people are becoming more noteworthy. Although the trustworthiness of these automated decisions is still up for debate, AI-centered technologies have been used extensively in a range of critical activities including the optimization of medical diagnosis [4], automated trading systems [28], hiring [19], online advertisement [36], immigration detention [23], welfare eligibility [11], etc. Although, these automated decision-making models enable efficient, data-driven and scalable decision making across many domains in society, they tend to be inherently biased against certain social groups since they learn from training data that preserves historical biases. Many fairness criteria have been proposed over the years to address this unwanted discrimination by machine learning models. In general many known fairness criteria revolving around algorithmic discrimination can roughly be divided into three categories: *independence*, *separation* and *sufficiency*. Most of the proposed fairness measures can be mathematically formalized as statements in terms of three random variables including sensitive attribute A, the target variable Y, and the classifier prediction output R. For example, the *independence* (such as statistical parity, demographic parity and group fairness) criterion requires the prediction score R to be statistically independent of the sensitive attribute A. *Separation* (such as equalized odds, equalized opportunity and conditional procedure accuracy) satisfies when the prediction output R is independent of the sensitive attribute A, provided that the desired output (target variable) is known. *Sufficiency* (such as conditional use accuracy, predictive parity, and calibration) requires that the desired output (target variable) is conditionally independent of the sensitive attribute A, given the system prediction output R. As compared to these abstract group-based fairness metrics, which compare scores across different groups, there are individual fairness metrics that focus on fair outcome for each participating individual. While each of these notions is appealing on its own, it has been shown that the main problem with these correlation-based fairness criteria is that they fail to detect social biases, especially in presence of statistical anomalies such as Simpson's paradox [34]. Furthermore, depending on the relationship between sensitive attributes and the data, particular notions of fairness may even lead to more biased decisions by computational algorithms [24]. Because of these limitations, it is now widely accepted that there is a need for more sociotechnical notions of fairness that are not based purely on probabilistic independence.

Several recent works have argued for the use of counterfactual and contrastive fairness criteria, which build on causal inferential models, and are related to both group and individual fairness concepts. Unlike statistical fairness approaches, these more recent notions of fairness are not merely based on data, instead, they consider additional knowledge about the structure of the world, in the form of a causal models. This additional information provides the possibility of evaluating the effect of interventions and understand how they propagate in a system.

Both counterfactual and contrastive fairness are connected with the core terms in reasoning and causal inference. Some researchers tend to collapse these two types of reasoning and treat them equivalently. However, there are good reasons to believe that contrastive reasoning should be distinguished from its counterfactual counterpart, and that they are equally important for a complete assessment of causation [18]. As the empirical studies show [10, 27], counterfactual and contrastive reasoning emphasizes on different aspects of causation and they differ in the dimension affected. Indeed, scholars from many disciplines have converged on the insight that a complementary contrastive counterfactual approach is needed to explain people's (and AI's) causal responsibility judgments [8, 14, 15, 33]. Despite their importance, there exist only a few studies that operationalize the concept of contrastive counterfactual causality to provide causal *explanation* for the automatically made predictions by ML/AI algorithms. Taking a different view, in this paper, we formalize probabilistic interpretation of contrastive counterfactual causality as essential feature to build *fair* causal models to encourage users' trust and to increase applicability of such automated systems. To the best of our knowledge, this is the first probabilistic fairness-aware augmentation approach with the contrastive counterfactual causality at its core. Our proposed bias mitigation approach uses a pre-processing mechanism to address fairness in algorithmic decision making, and for this reason it is very flexible, as it makes no assumptions about the choice of subsequently applied AI/ML algorithm, and it can be deployed with any automated modeling technique.

We adopt the standard definitions of necessary and sufficient causation to minimize the bias by augmenting the dataset with only necessary and sufficient number of data points, such that the overall dataset represents a more equitable world. Our data augmentation approach is **i)** **efficient** , as the level of augmentation is not arbitrarily decided by domain experts or policy makers; instead, it is decided based on theoretically-grounded probabilistic measures of necessity and sufficiency scores, which, respectively quantify the necessity and sufficiency of an attribute for an algorithm's decision [13]; and **ii)** **effective**, as it extensively reduces the bias, while accuracy is not significantly compromised.

Our data augmentation technique can be described using an example as follows. Consider a loan granting system dataset including the protected attribute gender, and assume that the dataset is biased against females. We create an "intervened dataset" by augmenting counterfactual examples of **i)** females whose loan applications were actually not approved, and the algorithm's decision was also a decline; however, in a counterfactual world, where the gender was changed from female to male, the algorithm's decision would have been approval; **ii)** males whose loan applications were actually approved, and the algorithm also predicted approval; however, algorithm's decision would have been negative had the gender was female.

We compared our model against two baselines on two benchmark datasets. The first baseline model is the one with no modification on the dataset (no mitigation), and second one is augmenting the dataset with only counterfactual examples. Our experimental results show that our augmentation technique significantly reduces bias in terms of three fairness criteria: accuracy parity, equalized odds parity, and demographic parity between protected and unprotected groups, while it keeps the accuracy nearly constant for two datasets.

2 BACKGROUND

Cambridge dictionary defines fairness as "the quality of treating people equally or in a way that is right or reasonable"[1]. According to [30], the complete consideration of algorithmic fairness for a society can be identified with three main components: **i)** the overarching social objective of a model deployment in a decision-making problem, **ii)** the set of individuals subject to the decision, and **iii)** the set of decisions that decision-makers could encounter while interacting with predictions generated by the model. This definition is adopted by the AI literature in different ways, and researchers brought different mathematical descriptions for the further use of fairness concepts in AI algorithms. Broadly speaking, the fairness problem from a computational point of view can be classified into two main categories: *disparate impact* and *disparate treatment*, which approaches the fair decision-making at a resolution of the group- and individual-level, respectively [39]. Recent studies also proposed hybrid measurements at a subgroup-level, which aim considering the group and individual notions of fairness, simultaneously.

Even though the concept of fairness is a highly desirable quality from a societal perspective, it can be extremely difficult to implement in practice, since there is no unique definition of algorithmic fairness to mitigate biases towards individuals, subgroups, or groups in utilization of decision-making algorithms. A very comprehensive taxonomy of different fairness measurements along with their definitions are given in [37]. Accordingly, fairness definitions are categorized into three main categories: Statistical, similarity-, and causal reasoning-based measures. Here, statistical measures cover definitions based on **i)** only predictive outcomes such as group fairness [9] as well as conditional statistical parity [7], **ii)** predicted and actual outcomes such as predictive parity [6], false-negative error rate balance [6], conditional use accuracy equality [2], overall accuracy equality [2], and treatment equality [2], and **iii)** predicted probabilities and actual outcome such as test fairness [6], well-calibration [22], balance for positive class [22] and balance

[1] https://dictionary.cambridge.org/us/dictionary/english/fairness

for negative class [22]. Furthermore, similarity-based measures such as causal discrimination [12], fairness through unawareness [24] and fairness through awareness [9] are proposed since the aforementioned statistical methods largely ignore the relationship between other attributes and sensitive attributes by marginalizing over insensitive attributes of the classified subject [37].

To capture the higher-order interactions of sensitive attribute(s) with others, new approaches have been introduced to particularly embed algorithmic fairness into the causal models. Following the notation of Pearl [32] and other related studies [1, 40, 41], given a probabilistic causal model, $\mathcal{M} = ((\mathcal{U}, \mathcal{V}, \mathcal{R}), \mathcal{F})$, the fairness in a causal relationship is investigated based on an observational approach. Here, $\mathcal{U}$ represents the set of exogenous variables $\mathcal{U} = \{U_1, U_2, ..., U_m\}$ and $\mathcal{V}$ represents the set of endogenous variables $\mathcal{V} = \{V_1, V_2, ..., V_n\}$. Moreover, $\mathcal{R} = \{R_1, R_2, ..., R_t\} = \{v_1, v_2, ..., v_n, ..., u_1, u_2, ..., u_m\}$ denotes a set of possible values for endogenous and exogenous variables. Furthermore, $\mathcal{F}$ is a set of functions $\{f_1, f_2, \ldots, f_n\}$, each defining an structural equation for the variables $\{V_1, V_2, ..., V_n\}$. Also, let Y be the expected outcome and $\hat{Y}$ be the predicted outcome.

Assume that the predictive function f is given by the structural equation of the endogenous variable $\hat{Y}$. Also, assume the partitioning of the remaining variables $\mathcal{U} \bigcup \mathcal{V} \backslash \{\hat{Y}\}$ into a set of protected attribute $\mathcal{A}$, corresponding to discriminatory elements of information and a set of features $\mathcal{X}$, carrying no sensitive information. The predictor $\hat{Y} = f(\mathcal{A}, \mathcal{X})$ is considered a fair model if it does not discriminate with respect to the protected attributes $\mathcal{A}$.

Although there are numerous definitions of fairness through causal reasoning, e.g., no unresolved discrimination [21], no proxy discrimination [21] and fair inference [31], the most commonly used notion is the *counterfactual fairness* [24] which formulates fairness as the equivalence of actual data and its counterfactual quantity. Recently, Zennaro et al. [40, 41] also proposed another algorithmic fairness concept to be applicable for causal inference frameworks, called *contrastive fairness.* Broadly speaking, counterfactual explanations aim to find an answer to the "what if" question, while contrastive frameworks focus on the question of "why this and not that?". The next section will explain these concepts in more detail.

3 THEORY

3.1 Contrastive and Counterfactual Explanations in Causal Judgment

Counterfactuals are one of the most commonly used explanatory phenomena in cognitive science in which it is assumed that the mind compares actual events with alternatives and constructs mental representations accordingly [3]. In fact, counterfactuals depict events and states of the world that did not take place and contradict global facts in either implicit or explicit ways [35]. Thus, a counterfactual can be considered as a conditional statement about an alternative possibility and its consequences of the form "What would be the outcome (if Y occurs or not) if P occurred instead of Q[17]?"

On the other hand, research in humanities and social science shows that explanation is intrinsically contrastive [29]. In contrastive reasoning, an explanation tries to find an answer to the why-question regarding the cause of the event in terms of hypothesized non-occurring outcome alternatives, i.e., "Why did Y happen rather than Z [25]?" Thus, many researchers argue that contrastive explanations have a necessary and sufficient capability of distinguishing answers to explanatory questions by using a large set of contrastive hypothesized alternatives that provide sufficient detailed information about the reasoning behind the question [35].

Counterfactual and contrastive explanations are oftentimes considered as similar concepts. For example, Lombrozo [26] states that both hypothesize the non-occurred cases compared to reality, and counterfactual reasoning is actually a subset of contrastive reasoning. Although both counterfactual and contrastive reasoning seem to be similar to each other (both consider instances in which some variables are negated to bring causal explanations), they differ in the dimensions affected. In counterfactual reasoning, the candidate is absent (e.g. would the employee have failed had she not been a woman?), and the emphasis is on the necessity of a factor; whereas, in contrastive reasoning, the effect is absent (e.g. what made the difference between the employee failed and the employees who did not fail?), thus the focus is on the sufficiency of an attribute [27].

3.2 Contrastive and Counterfactual Explanations in the Context of Fairness

In recent years, counterfactual reasoning is adopted to the fairness literature, it was first introduced by Kusner et al. [24]. Mathematically speaking, given the sensitive attribute A and the observed variable $\mathcal{X}$, a predictor $\hat{Y}$ is counterfactually fair, when it holds

$$\mathbb{P}(\hat{Y}_{A\leftarrow a}(U) = y|\mathcal{X} = x, A = a) = \mathbb{P}(\hat{Y}_{A\leftarrow a'}(U) = y|\mathcal{X} = x, A = a) \tag{1}$$

for all y and $a \neq a'$. This equation ensures that the distribution over possible predictions is the same in both the actual world and a counterfactual world where the sensitive attribute(s) were different while all other conditions are left unchanged.

Although counterfactual fairness captures the notion of fairness for people who belong to different groups, the criterion is applicable only on population-level. Societal categories often do not permit manipulation of counterfactuals, and thus fail to satisfy the demands for evaluating the fairness [20]. To overcome this challenge, Chakrabarti et al. proposed contrastive fairness in algorithmic decision-making [5], where they suggested three contrastive fairness definitions. The first one aims to answer the question of "Is it fair to make decision D for individual I, instead of decision D". Given the sensitive attribute A and observed variables $\mathcal{X}$ for a particular individual i, a predictor $\hat{Y}$ is *D-contrastive fair* if

$$\mathbb{P}(\hat{Y}_{A_i\leftarrow a}(U_i) = y|\mathcal{X}_i = x, A_i = a) = \tag{2}$$

$$\mathbb{P}(\hat{Y}_{A_i\leftarrow a'}(U_i) = y|\mathcal{X}_i = x, A_i = a) \tag{3}$$

In the second one, the fairness criteria is considered in terms of individuals. This definition aims to answer the question of "Is it fair to make decision D for individual I, but D' for individual J?" Accordingly, given the sensitive attribute A and observed variables $\mathcal{X}$ for a particular individual i, a predictor $\hat{Y}$ is *I-contrastive fair* if

$$\mathbb{P}(\hat{Y}_{A_i\leftarrow a_i}(U_i) = y|\mathcal{X}_i = x, A_i = a_i) = \tag{4}$$

$$\mathbb{P}(\hat{Y}_{A_i\leftarrow a_i'}(U_i) = y|\mathcal{X}_i = x, A_i = a_i) \tag{5}$$

Lastly, the third definition is to answer "Is it fair to make decision D for individual I at time t, but D' at time t'?". The main question here is, whether or not it is fair, if over time the decision made for an individual changes. Thus, given the sensitive attribute A and observed variables $\mathcal{X}$ for a particular individual i over all time points t, a predictor $\hat{Y}$ is *T-contrastive fair* if

$$\mathbb{P}(\hat{Y}_{A_i \leftarrow a_i}(U_i) = y | \mathcal{X}_i = x_i(t), A_i = a_i) = \tag{6}$$

$$\mathbb{P}(\hat{Y}_{A_i \leftarrow a'_i}(U_i) = y | \mathcal{X}_i = x_i(t), A_i = a_i) \tag{7}$$

3.3 Our Proposed Method: Contrastive Counterfactual Fairness

Unlike counterfactual fairness, there are not many studies that deploy contrastive fairness in the context of fair decision-making. Despite the promising aspects of contrastive reasoning in algorithmic fairness, these are the only definitions that are adopted from contrastive reasoning, and it is essentially the same as counterfactual fairness, but for a specific individual. These two reasoning techniques, however, are fundamentally different from each other. More importantly, they are complementary. This complementarity is pointed out with the definition of necessity and sufficiency scores defined in [13], which can be translated into notions as below:

$$NEC_a^{a'}(U) = \mathbb{P}(\hat{Y}_{A \leftarrow a'}(U) = y' | \mathcal{X} = x, A = a, Y = y) \tag{8}$$

$$SUF_a^{a'}(U) = \mathbb{P}(\hat{Y}_{A \leftarrow a}(U) = y | \mathcal{X} = x, A = a', Y = y') \tag{9}$$

Here, the first notion (Equation 8) can be considered as follows: in the case of a negative outcome, the probability of algorithm's decision to be positive instead of negative had A been a instead of a'. In the second one (Equation 9), when the outcome is positive, the probability that for individuals with attributes x, the algorithm's decision would be negative instead of positive had A been a' instead of a. Since these measures are introduced for the explainability of black-box AI models but we aim to translate it into fairness point of view, we present a methodology that combines these two phenomena by considering the individuality concern of contrastive fairness. A predictor $\hat{Y}$ is ***contrastive counterfactual fair*** given the sensitive attribute A and observed variables $\mathcal{X}$ for a particular individual i if

$$\mathbb{P}(\hat{Y}_{A_i \leftarrow a}(U) = y | \mathcal{X}_i = x, A_i = a, Y_i = y) \tag{10}$$

$$= \mathbb{P}(\hat{Y}_{A_i \leftarrow a'}(U) = y | \mathcal{X}_i = x, A_i = a, Y_i = y) \tag{11}$$

and

$$\mathbb{P}(\hat{Y}_{A_i \leftarrow a}(U) = y | \mathcal{X}_i = x, A_i = a', Y_i = y') \tag{12}$$

$$= \mathbb{P}(\hat{Y}_{A_i \leftarrow a'}(U) = y | \mathcal{X}_i = x, A_i = a', Y_i = y') \tag{13}$$

If these equations are satisfied then one can surmise that the contrast counterfactual in the decision made between these two individuals belonging to different sub-populations is fair. On the contrary of common metrics that only try to either false-positive rates, true positive rates or both for privileged and unprivileged groups, it guarantees that a distribution over contrastive predictions for an individual should remain unchanged in a world where an individual's sensitive attributes had been counterfactually different in a causal sense. In other words, changing sensitive attribute while keeping other attributes constant should not change the distribution of outcome variable conditioned on its true value.

3.4 Contrastive Counterfactual Fairness as a Data Augmentation Technique

	ORIGINAL DATA				COUNTERFACTUAL DATA			
Decision	Sex	Purpose	Housing	Prediction	Sex	Purpose	Housing	Prediction
1	female	car	rent	1	male	car	rent	0
1	male	house	rent	1	female	house	rent	0
0	male	education	own	0	female	education	own	0
1	female	car	rent	1	male	car	rent	0
0	female	car	own	0	male	car	own	1
0	male	education	own	0	female	education	own	0
0	female	car	own	0	male	car	own	1
1	male	house	rent	1	female	house	rent	1
1	male	education	rent	0	female	education	rent	1
0	female	education	own	0	male	education	own	1

Figure 1: An illustrative example of contrastive counterfactual fairness as a data augmentation technique.

To show how our data augmentation technique works to achieve contrastive counterfactual fairness in algorithmic decision-making, consider a hypothetical example, shown in Figure 1, illustrating an example of loan approval decision, where the protected attribute is the applicant's gender. The goal is to create an "intervened dataset" wherein we augment the original dataset with contrastive counterfactual examples, leading to more fair decisions made by the algorithm. In order to build our "intervened dataset", first, we create counterfactual data points by converting only the gender attribute of all applicants from their actual values to counterfactual values, while all other attributes are kept unchanged (right hand side table in Figure 1).

ORIGINAL DATA				INTERVENED DATA			
Sex	Purpose	Housing	Decision	Sex	Purpose	Housing	Prediction
female	car	rent	1	female	car	rent	1
male	house	rent	1	male	house	rent	1
male	education	own	0	male	education	own	0
female	car	rent	1	female	car	rent	1
female	car	own	0	female	car	own	0
male	education	own	0	male	education	own	0
female	car	own	0	female	car	own	0
male	house	rent	1	male	house	rent	1
male	education	rent	1	male	education	rent	1
female	education	own	0	female	education	own	0
				female	house	rent	1
				male	car	own	0
				male	car	own	0
				male	education	own	0

Figure 2: An intervened dataset created by augmenting the original dataset with contrastive counterfactual examples

We, then, select only the contrastive examples from these counterfactual data, based on sufficiency and necessity criteria (see their definitions in Section 3.3). Using sufficiency metric, we selected unprivileged individuals whose actual and predicted outcomes were negative, but the algorithm's decision would have been positive if those individuals belonged to the privileged group. In our example, this corresponds to females whose actual and predicted decisions were 0, but the algorithm's decision would have been 1 if the sex was male instead of female (examples in red), i.e., the sufficiency score 1 = $3/5 = 0.6$. Furthermore, using the necessity metric, we selected privileged individuals whose actual and predicted outcomes were positive, but the algorithm's decision would have been negative if those individuals belonged to the unprivileged group. In our example, this case is represented by males whose actual and predicted decision were 1, but the algorithm would have predicted them as 0 if the sex was female instead of male (examples in blue), i.e., the necessity score = $1/5 = 0.2$. Then, we create an "intervened dataset" by augmenting these contrastive counterfactual examples (colored in green) to the original dataset, as shown in Figure 2.

4 EXPERIMENTS

4.1 Datasets and Pre-processing

4.1.1 Adult Data Set. This dataset[2] is extracted from "1994 Census database" and contains 48,842 entries each describing one individual with 14 attributes, which are the age of the individual, his employment status, an education level (both categorical and numeric), marital status, occupation, race, biological sex, native country, capital gain and capital loss along with the hours an individual has reported working per week, the relationship that represents what this individual is relative to others and the final weight represents the estimated number of units in the target population. The task is to predict if an individual's annual income exceeds 50,000 dollars based on census data. The data set contains a distribution of 23.93% entries labeled with higher income (i.e., $> 50,000$) and 76.07% entries labeled with lower income status (i.e., $\leq 50,000$). Since the ratio of female individuals with lower income compared to high-income level is much higher than that of male individuals (Table 1), most of the classification algorithms are found to be gender-biased.

Table 1: Sex vs. Income Level in UCI Adult Dataset

	Low income	High income
Female	14,423	1,769
Male	22,732	9,918

For the pre-processing, the categorical education variable is not considered since the numerical variable of education level gives the same information. Furthermore, both categorical variables are encoded as binary or one-hot variables. E.g., native country, marital status, capital gain, and capital loss are represented as binary variables as the US vs non-US, single vs couple, higher or lower than zero respectively; whereas, the rest of the categorical variables such as relationship, race, work class, and occupation are identified as a one-hot variable. Furthermore, the numerical variables are normalized. In the prediction tasks, we split the dataset into 80% training set and 20% testing.

[2]https://archive.ics.uci.edu/ml/datasets/adult

4.1.2 German Credit Dataset. This publicly available dataset[3] contains 1,000 entries of loan applications of which prediction tasks aim to decide whether the application should be approved or not based on the applicant's profile. The bank evaluates two types of risk when it makes its decision: A good credit risk applicant is likely to pay back his loan, whereas a bad credit risk applicant is unlikely to do so. As independent variables, the data set consists of 20 attributes recording the personal information and financial history of applicants. Among them, credit duration in a month, credit amount, installment rate in percentage of disposable income, number of existing credits, number of dependent people, and moving year of present residency are represented with numerical variables; while the status of existing checking account, credit history, purpose, savings account/bonds, employment status, personal status and sex, other debtors/guarantors, property, other installment plans, housing, job status, phone registration, and foreign worker status are categorical variables.

Table 2: Foreign Worker Status vs. Success of Loan Application in German Credit Dataset

	Bad application	Good application
Foreign worker	4	33
Domestic worker	296	667

The data set contains a distribution of 30% entries labeled as bad application and 70% entries labeled as a good application. Since the likelihood of a foreign worker having a good application is much higher than that of a domestic worker (Table 2), most of the classification algorithms are found to be biased. For the pre-processing, all categorical variables are encoded as a one-hot variable, and numerical values are normalized. In the prediction tasks, we split the dataset into 80% training set and 20% testing.

4.2 Performance Metrics

As in the determination of an algorithm's performance in a classification problem, most of the fairness metrics can be also identified based on the metrics of a confusion matrix. Thus, we'll start with the definitions of confusion matrix elements to better explain the performance metrics used in this study.

- *True positive (TP)* is the case where the model correctly predicts the positive class.
- *False positive (FP)* is the case where predicted outcome assigned to be positive class while the actual outcome belongs to the negative class.
- *True negative (TN)* is the case where the model correctly predicts the negative class.
- *False negative (FN)* is the case where predicted outcome assigned to be negative class while the actual outcome belongs to the positive class.

4.2.1 Accuracy. This is one of the most common metrics to measure the performance of a classification algorithm, which summarizes the performance of a classification model as the number of

[3]https://archive.ics.uci.edu/ml/datasets/statlog+(german+credit+data)

correct predictions divided by the total number of predictions. In terms of confusion matrix elements, we calculated accuracy as $(TP + TN)/(TP + FP + TN + FN)$.

4.2.2 Accuracy Parity. This measure [38] aims to equate accuracy values across the sensitive groups, i.e. Accuracy parity emphasizes the equality of both true positive and true negative rates, and calculated as follows:

$$\mathbb{P}(\hat{Y}_{A\leftarrow a} = 1|A = a, Y = 1) = \mathbb{P}(\hat{Y}_{A\leftarrow a'} = 1|A = a', Y = 1) \quad (14)$$

$$\mathbb{P}(\hat{Y}_{A\leftarrow a} = 0|A = a, Y = 0) = \mathbb{P}(\hat{Y}_{A\leftarrow a'} = 0|A = a', Y = 0) \quad (15)$$

In terms of confusion matrix elements, accuracy parity is calculated by subtracting the accuracy value of protected group from that of unprotected group.

4.2.3 Demographic Parity. This measure [9, 37], which is also commonly referred to as statistical parity, is designed to mathematically represent the legal concept of disparate impact by ensuring the likelihood of the equality of a positive outcome across protected and unprotected groups. A predictor $(\hat{Y})$ satisfies demographic parity if

$$\mathbb{P}(\hat{Y}_{A\leftarrow a} = 1|A = a) = \mathbb{P}(\hat{Y}_{A\leftarrow a'} = 1|A = a') \quad (16)$$

In terms of confusion matrix elements, demographic parity is calculated by subtracting the $TP + FP$ value of protected group from that of unprotected group.

4.2.4 Equalized Odds Parity. The measure [16] aims to assign equal probability to either true positive predictions or false-positive rates between different groups, and calculated as follows: A predictor $(\hat{Y})$ satisfies the quality of opportunity if

$$\mathbb{P}(\hat{Y}_{A\leftarrow a} = 1|A = a, Y = 1) = \mathbb{P}(\hat{Y}_{A\leftarrow a'} = 1|A = a', Y = 1) \quad (17)$$

$$\mathbb{P}(\hat{Y}_{A\leftarrow a} = 1|A = a, Y = 0) = \mathbb{P}(\hat{Y}_{A\leftarrow a'} = 1|A = a', Y = 0) \quad (18)$$

In terms of confusion matrix elements, demographic parity is calculated by subtracting the $\frac{TP}{TP+FN}$ value of the protected group from that of the unprotected group.

5 RESULTS

In this study, we propose a new algorithmic fairness definition called *contrastive counterfactual fairness* (see Section 2.4) that integrates two commonly-used reasoning techniques in the explanatory analysis of AI decision-making, i.e. counterfactual and contrastive reasoning. These two approaches are believed to be complementary to each other and one can sufficiently explain detailed information about the reasoning behind the question by considering the large set of contrastive hypothesized alternatives (counterfactuals). Since contrastive counterfactual reasoning is extensively adopted in explainable AI literature; thus, we believed that this definition satisfies the complete fair treatment between two individuals

To test the efficiency of the proposed fairness definition, we carried out experiments on two mostly-used fairness-related datasets, i.e., UCI Adult and German credit datasets (see Section 2.5) by applying a data augmentation technique to increase the fairness of the decision-making algorithm for both protected and unprotected groups. As prediction algorithms, we deployed logistic regression (LR) with limited-memory BFGS optimization algorithm, multilayer perceptron (MLP), and support vector machine (SVM) classifiers. Despite the existence of multiple sensitive attributes in both datasets, we selected one sensitive attribute in the experimental analysis for the sake of simplicity. For the performance criteria, we not only measured the changes in the accuracy of the prediction analysis but also computed fairness parity measures such as accuracy, equalized odds, and demographic parity (see Section 2.6). Therefore, we can easily argue that the final results will yield the efficiency of the proposed results in terms of the accuracy-fairness trade-off.

Table 3: Accuracy of Prediction Algorithms using UCI Adult Dataset

Method	LR	MLP	SVM
No mitigation	.7633	7775	.7598
Counterfactual	.7511	.7545	.7733
Contrastive counterfactual	.7539	.7604	.7721

In the Adult dataset, we aim to fairly classify if an individual's annual income exceeds 50,000 dollars based on census data by debiasing the outcome with respect to sex attributes. Despite the main concern of improving the fairness of the classification algorithm, we also target to not compromise accuracy. Therefore, we first computed accuracy values of three different classification algorithms; first, when there is no modification in the dataset and classification method (no mitigation), second, when counterfactual data of sex attribute is augmented (counterfactual) before classification, and third, when counterfactual data of contrastive examples is augmented (contrastive counterfactual). Although the additional fairness criteria in the optimization of a classification strategy impair its accuracy of it, the counterfactual data augmentation technique seems to not harm the performance of the overall performance significantly. In the deployment of the SVM classifier, even higher accuracy values are observed. Contrastive counterfactual data augmentation, on the other hand, slightly yielded better results than counterfactual data augmentation in the deployment of LR and MLP. These results show that our proposed strategy does not significantly compromise the accuracy of the classification algorithm while trying to mitigate the bias in its deployment. Furthermore, the German Credit dataset also provided similar results.

For comparing and contrasting the fairness performance of the proposed fairness definition; we calculated accuracy, equalized odds, and demographic parity scores of LR, MLP, and SVM classification algorithms when there is no mitigation, with only counterfactual, and contrastive counterfactual data augmentation techniques (Figure 3). In the case of the UCI Adult dataset; since algorithms tend to falsely assign lower-income values for females compared to males but predict true values similarly across different sex attributes, true predictions (both TP and TN) are not much affected. Therefore, accuracy parity results show different patterns in the utilization of bias mitigation techniques. While accuracy parity is decreased, i.e., similar accuracy values are obtained for individuals being in both protected and unprotected groups, in the utilization of counterfactual data augmentation before LR deployment, the value increased in the application of MLP and SVM. Contrastive counterfactual data augmentation, on the other hand, yielded better results with SVM, but less optimal outcomes with LR and MLP. Furthermore, in the

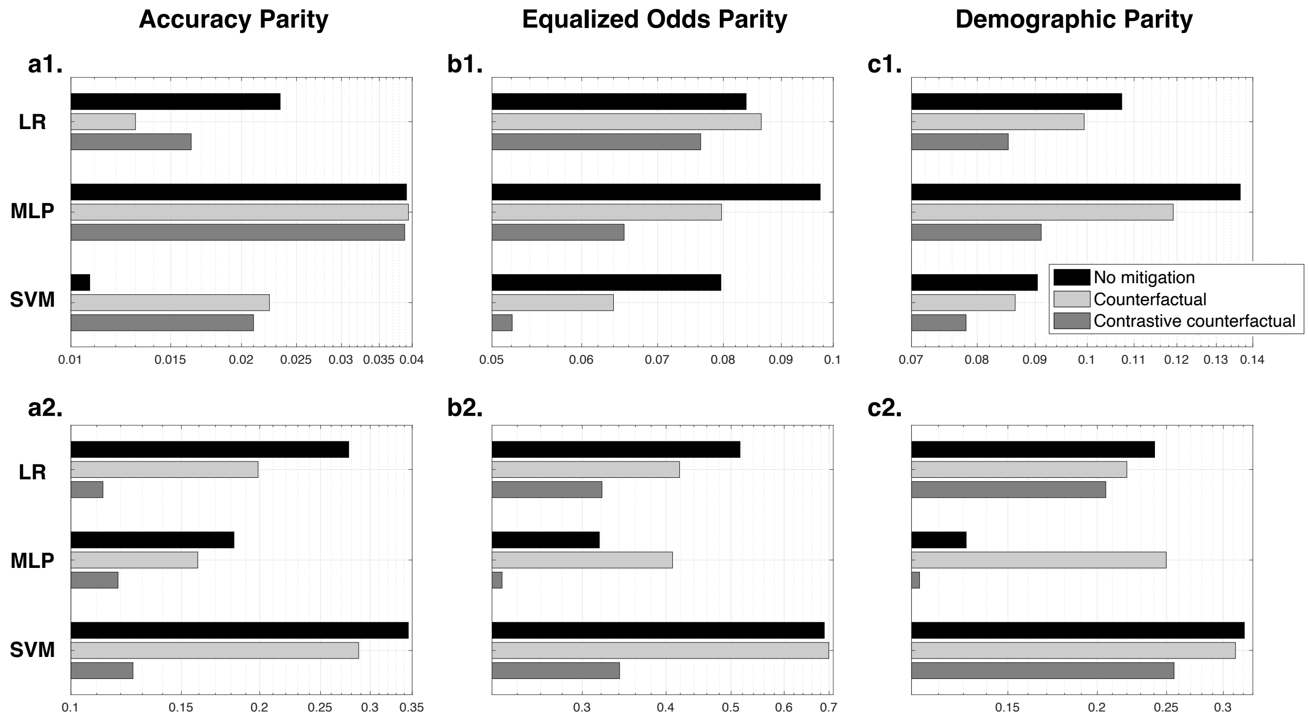

Figure 3: Fairness performance metrics (a. Accuracy parity, b. Equalized odds parity, c. Demographic parity) in the deployment of logistic regression (LR) with limited-memory BFGS optimization algorithm, multi-layer perceptron (MLP) and support vector machine (SVM) classifiers on 1. UCI Adult (top), 2. German Credit datasets (bottom). We compared our results with a baseline model that is the one with no modification on the dataset (no mitigation) and a second one in which all counterfactual data w.r.t sensitive attribute is augmented (counterfactual). Our experimental results show that our contrastive counterfactual augmenting technique significantly reduces bias in terms of three fairness criteria: accuracy parity (in most cases), equalized odds parity, and demographic parity between protected and unprotected groups, while it keeps the accuracy nearly constant.

German credit dataset, the best performances are obtained with contrastive counterfactual data augmentation before all classification algorithms and followed by counterfactual data augmentation.

When it comes to equalized odd parity, since it calculates the differences of $TP + TN$ values for female and male individuals, counterfactual data augmentation does not yield better results in all cases. For example, it gives more favorable parity scores in the application of MLP and SVM in the UCI Adult dataset and that of LR and SVM in the German credit dataset compared to the case in which there is no bias mitigation. Whereas, contrastive counterfactual data augmentation yields the best performances with the lowest equalized odd parity scores in the deployment of all three classification techniques on both datasets. In demographic parity comparison, we also recognize similar results. Contrastive counterfactual data augmentation yields the best results on both datasets regardless of the classification technique deployed.

We believe that this can be explained as follows: Counterfactual data augmentation simply duplicates data and convert the sex attribute to prevent biased outcomes toward this sensitive attribute. However, true and false prediction ratios are not the same for females and males, and adding all counterfactual data will change the direction of the bias, but will not extinguish completely. On the other hand, contrastive counterfactual data augmentation provides a more logical data pre-processing since counterfactuals (data which of sex attribute is converted only) for contrastive examples are augmented only. In this technique, the necessity score in Equation 8 measures the algorithm's percentage of positive decisions that are attributable to or due to the attribute value x, while the sufficiency score in Equation 9 relates that "What would be the probability that for individuals with attributes x, the algorithm's decision would be positive instead of negative had A been a instead of a'?". In contrastive counterfactual data augmentation, we concatenated counterfactual examples of **i)** females if the outcome is actually negative, and prediction algorithm results with the negative outcome; however, the algorithm would predict positive if the individual is a male, and **ii)** males if the outcome is positive, and prediction algorithm results with a positive outcome; however, the algorithm would predict negative if the individual is a female. This selective contrastive counterfactual data augmentation yield a more fairly populated dataset in pre-processing bias mitigation.

6 DISCUSSION

The impressive performances of AI algorithms have recently been challenged after studies suggested that some of these algorithms are biased and might produce unfair outcomes for some minority groups. In order to mitigate bias in algorithmic decision-making, researchers proposed fair causal learning approaches, which facilitates modeling the cause and effect knowledge structure, allowing to discover the source of bias, and thus, improving transparency and explainability of AI algorithms. Aligned with this line of research, we proposed a pre-processing method that modifies the sample distribution of protected attributes using a probabilistic contrastive counterfactual data augmentation approach, which minimizes the bias of the algorithm's decision, both effectively and efficiently. Similar to the work in [13], our proposed methodology is rooted in theoretically-grounded models of causal judgment. However, our work differs from the methodology in [13], where they define and use the necessity and sufficiency scores to generate post-hoc *explanations* for black-box decision-making algorithms. Our approach, on the other hand, utilizes the necessity and sufficiency scores to create a "repaired" dataset during the pre-processing, and that can be deployed by AI algorithms to address *fairness*. Our proposed method can mitigate bias at individual-level, and is model-agnostic, that is, it can be applied to any automated decision-making algorithms, since it does not make any assumptions about the choice of the modeling technique being applied. Our data augmentation technique can be described as follows: Given a dataset that contains a protected attribute, we first create an "counterfactual dataset" in which only protected attribute is converted while all other variables are kept same. Then, only the contrastive examples are selected from the counterfactual dataset in consideration of sufficiency and necessity scores. The former is considered by selecting the counterfactual data of unprivileged individuals whose outcome is actually negative, but the decision-making algorithm would predict it positive if the individual is belong to the privileged group. For the latter, counterfactual data of privileged individuals whose outcome is actually positive, but the decision-making algorithm would predict it negative if the individual is belong to the unprivileged group is selected. Then, we create an "intervened dataset" by augmenting these contrastive counterfactual examples. We compared our model against two baselines on two benchmark datasets which are UCI Adult and German credit datasets. The first baseline model is the one with no modification on the dataset (no mitigation), and second one is augmenting the dataset with only counterfactual examples. Our experimental results show that our augmenting technique significantly reduces bias in terms of three fairness criteria: accuracy parity, equalized odds parity, and demographic parity between protected and unprotected groups, while it keeps the accuracy nearly constant.

REFERENCES

[1] Dalal Alrajeh, Hana Chockler, and Joseph Y Halpern. 2020. Combining experts' causal judgments. *Artificial Intelligence* 288 (2020), 103355.

[2] Richard Berk, Hoda Heidari, Shahin Jabbari, Michael Kearns, and Aaron Roth. 2021. Fairness in criminal justice risk assessments: The state of the art. *Sociological Methods & Research* 50, 1 (2021), 3–44.

[3] Ruth MJ Byrne. 2016. Counterfactual thought. *Annual review of psychology* 67 (2016), 135–157.

[4] Longbing Cao, Chengqi Zhang, Thorsten Joachims, Geoffrey Ian Bawtree Webb, Dragos Margineantu, and Graham Williams. 2015. *KDD'15: Proceedings of the 21st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. Association for Computing Machinery (ACM).

[5] Tapabrata Chakraborti, Arijit Patra, and J Alison Noble. 2020. Contrastive fairness in machine learning. *IEEE Letters of the Computer Society* 3, 2 (2020), 38–41.

[6] Alexandra Chouldechova. 2017. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big data* 5, 2 (2017), 153–163.

[7] Sam Corbett-Davies, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. 2017. Algorithmic decision making and the cost of fairness. In *Proceedings of the 23rd acm sigkdd international conference on knowledge discovery and data mining*. 797–806.

[8] Maartje MA De Graaf and Bertram F Malle. 2017. How people explain action (and autonomous intelligent systems should too). In *2017 AAAI Fall Symposium Series*.

[9] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. 2012. Fairness through awareness. In *Proceedings of the 3rd innovations in theoretical computer science conference*. 214–226.

[10] Hillel J Einhorn and Robin M Hogarth. 1986. Judging probable cause. *Psychological Bulletin* 99, 1 (1986), 3.

[11] Virginia Eubanks. 2018. *Automating inequality: How high-tech tools profile, police, and punish the poor.* St. Martin's Press.

[12] Sainyam Galhotra, Yuriy Brun, and Alexandra Meliou. 2017. Fairness testing: testing software for discrimination. In *Proceedings of the 2017 11th Joint meeting on foundations of software engineering*. 498–510.

[13] Sainyam Galhotra, Romila Pradhan, and Babak Salimi. 2021. Explaining black-box algorithms using probabilistic contrastive counterfactuals. In *Proceedings of the 2021 International Conference on Management of Data*. 577–590.

[14] Tobias Gerstenberg, Noah D Goodman, David A Lagnado, and Joshua B Tenenbaum. 2015. How, whether, why: Causal judgments as counterfactual contrasts.. In *CogSci*.

[15] Eric Grynaviski. 2013. Contrasts, counterfactuals, and causes. *European Journal of International Relations* 19, 4 (2013), 823–846.

[16] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of opportunity in supervised learning. *Advances in neural information processing systems* 29 (2016), 3315–3323.

[17] Ph D Noel Hendrickson. 2011. *Counterfactual reasoning: A basic guide for analysts, strategists, and decision makers.* Lulu. com.

[18] Denis J Hilton. 1990. Conversational processes and causal explanation. *Psychological Bulletin* 107, 1 (1990), 65.

[19] Lily Hu and Yiling Chen. 2018. A short-term intervention for long-term fairness in the labor market. In *Proceedings of the 2018 World Wide Web Conference*. 1389–1398.

[20] Atoosa Kasirzadeh and Andrew Smart. 2021. The use and misuse of counterfactuals in ethical machine learning. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. 228–236.

[21] Niki Kilbertus, Mateo Rojas-Carulla, Giambattista Parascandolo, Moritz Hardt, Dominik Janzing, and Bernhard Schölkopf. 2017. Avoiding discrimination through causal reasoning. *arXiv preprint arXiv:1706.02744* (2017).

[22] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. 2016. Inherent trade-offs in the fair determination of risk scores. *arXiv preprint arXiv:1609.05807* (2016).

[23] Robert Koulish. 2017. Immigration detention in the risk classification assessment era. *Conn. Pub. Int. LJ* 16 (2017), 1.

[24] Matt J Kusner, Joshua R Loftus, Chris Russell, and Ricardo Silva. 2017. Counterfactual fairness. *arXiv preprint arXiv:1703.06856* (2017).

[25] Peter Lipton. 1990. Contrastive explanation. *Royal Institute of Philosophy Supplements* 27 (1990), 247–266.

[26] Tania Lombrozo. 2012. Explanation and abductive inference. (2012).

[27] Ann L McGill and Jill G Klein. 1993. Contrastive and counterfactual reasoning in causal judgment. *Journal of Personality and Social Psychology* 64, 6 (1993), 897.

[28] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2019. A survey on bias and fairness in machine learning. *arXiv preprint arXiv:1908.09635* (2019).

[29] Tim Miller. 2019. Explanation in artificial intelligence: Insights from the social sciences. *Artificial intelligence* 267 (2019), 1–38.

[30] Shira Mitchell, Eric Potash, Solon Barocas, Alexander D'Amour, and Kristian Lum. 2021. Algorithmic fairness: Choices, assumptions, and definitions. *Annual Review of Statistics and Its Application* 8 (2021), 141–163.

[31] Razieh Nabi and Ilya Shpitser. 2018. Fair inference on outcomes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.

[32] Judea Pearl. 2009. *Causality.* Cambridge university press.

[33] Jonathan Schaffer. 2005. Contrastive knowledge. *Oxford studies in epistemology* 1 (2005), 235–271.

[34] Edward H Simpson. 1951. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B (Methodological)* 13, 2 (1951), 238–241.

[35] Ilia Stepin, Jose M Alonso, Alejandro Catala, and Martín Pereira-Fariña. 2021. A survey of contrastive and counterfactual explanation generation methods for explainable artificial intelligence. *IEEE Access* 9 (2021), 11974–12001.

[36] Latanya Sweeney. 2013. Discrimination in online ad delivery: Google ads, black names and white names, racial discrimination, and click advertising. *Queue* 11, 3 (2013), 10–29.

[37] Sahil Verma and Julia Rubin. 2018. Fairness definitions explained. In *2018 ieee/acm international workshop on software fairness (fairware)*. IEEE, 1–7.

[38] Mingyang Wan, Daochen Zha, Ninghao Liu, and Na Zou. 2021. Modeling Techniques for Machine Learning Fairness: A Survey. *arXiv preprint arXiv:2111.03015* (2021).

[39] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez Rodriguez, and Krishna P Gummadi. 2017. Fairness beyond disparate treatment & disparate impact: Learning classification without disparate mistreatment. In *Proceedings of the 26th international conference on world wide web*. 1171–1180.

[40] Fabio Massimo Zennaro and Magdalena Ivanovska. 2018. Counterfactually fair prediction using multiple causal models. In *European Conference on Multi-Agent Systems*. Springer, 249–266.

[41] Fabio Massimo Zennaro and Magdalena Ivanovska. 2018. Pooling of causal models under counterfactual fairness via causal judgement aggregation. *arXiv preprint arXiv:1805.09866* (2018).